

# Noori

Computational Modeling of Quranic Semantics Through Arabic NLP and Multi-Stream Embedding Systems



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## Introduction

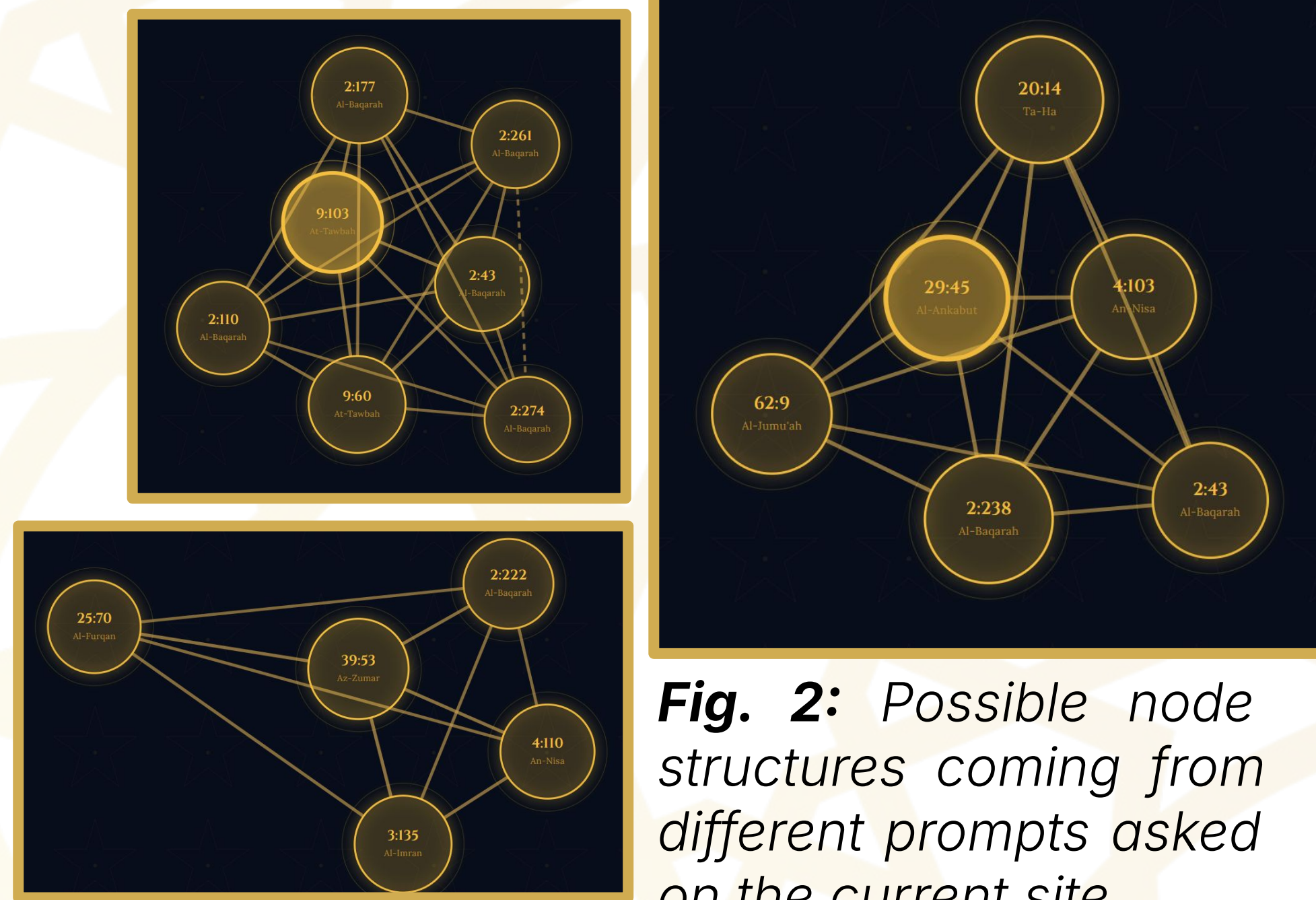
Most existing approaches to analyzing the Quran computationally rely on external sources such as tafsir (commentary from Islamic scholars) to understand meaning. While this is useful, it makes learning dependent on inconsistent data and shifts attention away from the Quran's own internal structure and beauty shown in **Fig. 2**. This project explores a different system, one that learns the meaning of verses using only the Quran itself.

## Novelty

Most work in Quranic study methods depends on external translation or basic keyword matching. In contrast, this system is designed to learn directly from the Quran's own internal relationships. Rather than taking meaning from outside commentary, the model finds connections (**Fig. 1**) through repeated phrases, shared roots, narrative patterns, and structural patterns. This avoids bias, and allows the user to interact with the entire Quran.

| Top 20 verses related to MERCY: |              |                                 |  |
|---------------------------------|--------------|---------------------------------|--|
| 1. 73:20                        | score: 0.898 | roots: rHm, gfr                 |  |
| 2. 2:109                        | score: 0.898 | roots: EfW                      |  |
| 3. 9:99                         | score: 0.898 | roots: rHm, gfr                 |  |
| 4. 3:19                         | score: 0.890 | roots: related (no direct root) |  |
| 5. 4:94                         | score: 0.890 | roots: related (no direct root) |  |

**Fig. 1:** Shows top 5 from 20 verses that relate to the words for mercy in arabic



**Fig. 2:** Possible node structures coming from different prompts asked on the current site

## Methods

Noori builds verse embeddings for the Quran by combining several feature extraction devices: root co-occurrence through PPMI and SVD, word structure through FastText, verse relationships through Node2Vec, and linguistic features from the Quranic Arabic Corpus.

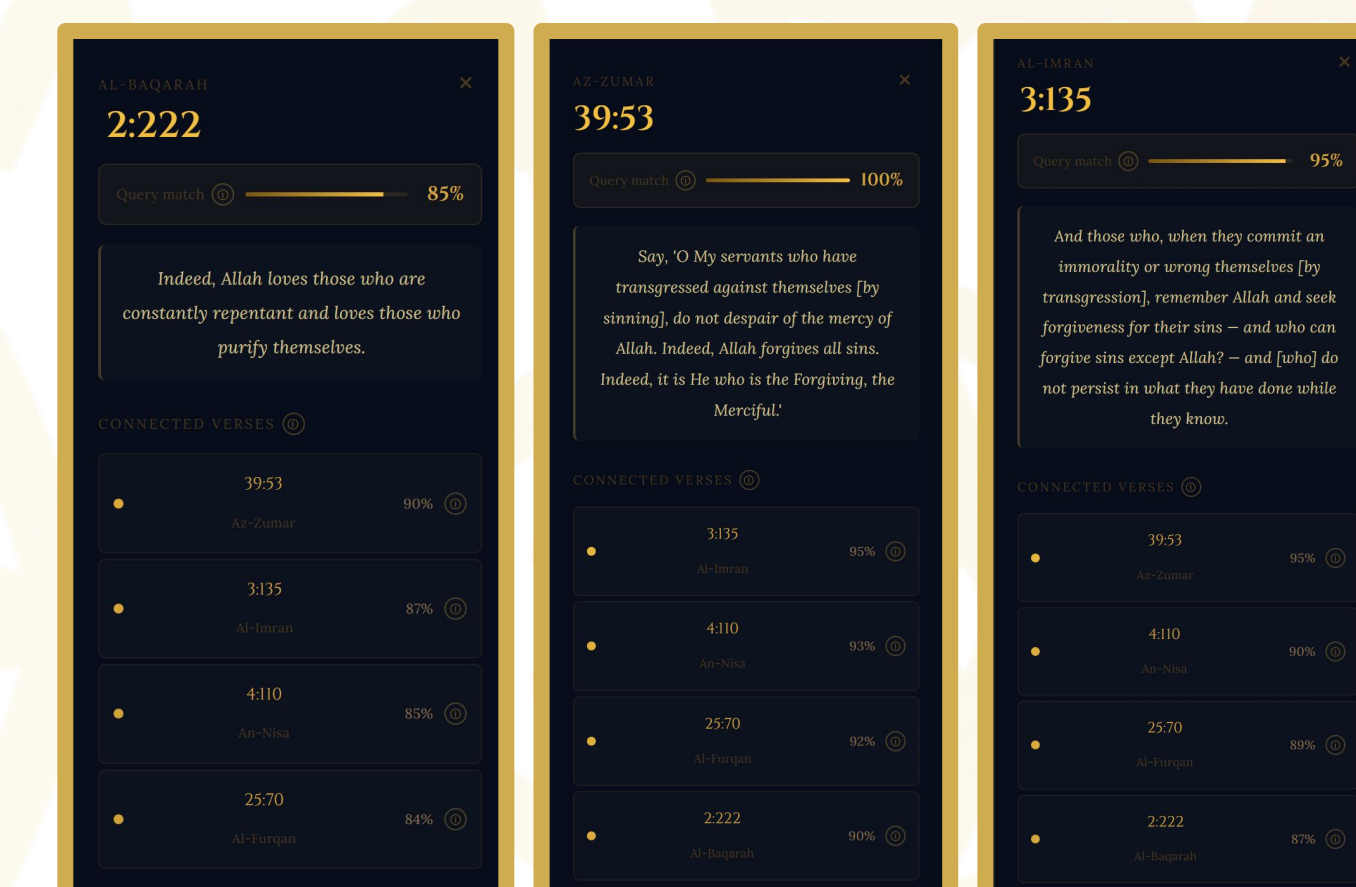
These devices are fused with a shared projection system and refined using an Attention layer and triplet loss, producing a semantic embedding matrix that supports the original structure intended with the project. By combining the extracted verses features into one vector per verse, it essentially creates a relative definition for each verse in our system (**Fig. 1**)

## Results

The system successfully generated a shared semantic embedding space for all 6,236 Quranic verses using only internal relationships. Related verses clustered together even when they did not share direct keywords, showing that the model captured conceptual similarity beyond simple text matching. The final embedding matrix supports the backend of the site, which allows the user to relate any queries to actual verses (**Fig. 3**) in the Quran.

## Future Work

Future development will focus on improving both the model and the educational experience built around it. Current plans on the backend include refining the training and improving the quality, as well as constantly updating the algorithms. I hope to turn Noori into a fully interactive app (**Fig. 2**) that allows users to explore the Quran.



**Fig. 3:** Verse analysis feature open on current site, allows the user to understand the verse node selected